

Surrogate Key Justification

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Surrogate Key Justification for STREET entity:

A surrogate key, `global_street_id`, was added to the STREET relation to avoid key bloat in its child tables. Although the natural key (`street_id`, `lcaut_name`) only consists of two attributes, STREET serves as a parent to the PROPERTY relation. Using the composite natural key would propagate to PROPERTY, resulting in a three-attribute foreign key. Introducing `global_street_id` simplifies these references, while the original composite key is still enforced through a candidate key (unique constraint).

Surrogate Key Justification for PROPERTY entity:

A surrogate key, `property_id`, was introduced to the PROPERTY relation. In the conceptual model, PROPERTY was a weak entity identified by a three-attribute composite primary key (`prop_street_no`, `street_id`, `lcaut_name`). Since PROPERTY serves as a parent to multiple relations, such as PROPERTY_OWNERSHIP and PHYSICAL_BIN, propagating this composite key would require child tables to store a three-attribute foreign key. The introduction of `property_id` simplifies foreign key references, while the original composite key is preserved through a candidate key (unique constraint) to enforce business rules.

Surrogate Key Justification for CONTRACT_WASTE entity:

A surrogate key, `ctwaste_id`, was introduced to the CONTRACT_WASTE relation to address a structural issue. Both CONTRACT_WASTE and BIN_CATALOGUE inherit `waste_type_id` from WASTE_TYPE, and propagating their natural composite keys into their shared child, CONTRACT_BIN_PRICING, resulted in overlapping foreign keys and duplication (e.g., redundant attributes like `waste_type_id1`). Introducing `ctwaste_id` enables a clean, non-identifying relationship that avoids this

duplication, while the original composite key (contract_no, waste_type_id) is preserved through a candidate key (unique constraint) to maintain business rules.

Surrogate Key Justification for BIN_CATALOGUE entity:

Similar to CONTRACT_WASTE, a surrogate key, binctl_id, was introduced to the BIN_CATALOGUE relation to resolve the diamond inheritance overlap. If the natural composite key (binctl_size, waste_type_id) were propagated to CONTRACT_BIN_PRICING alongside the key from CONTRACT_WASTE, it would create a foreign key collision. The use of binctl_id provides a clean, single-attribute reference that eliminates this conflict, while the original composite key is preserved through a candidate key (unique constraint) to enforce business rules.

Surrogate Key Justification for CONTRACT_BIN_PRICING entity:

A surrogate key, ctbinpri_id, was introduced to the CONTRACT_BIN_PRICING relation. In the normalized model, this relation was originally identified by a three-attribute composite primary key (contract_no, waste_type_id, binctl_size). As CONTRACT_BIN_PRICING serves as a parent to the PHYSICAL_BIN relation, propagating this composite key would require a three-attribute foreign key in the child table. The introduction of ctbinpri_id simplifies foreign key references, while the original composite key is preserved through a candidate key (unique constraint) to maintain business rules.

Surrogate Key Justification for COLLECTION_RUN entity:

A surrogate key, collrun_id, was introduced to the COLLECTION_RUN relation. Originally, this relation was identified by a three-attribute composite primary key (collrun_date, contract_no, waste_type_id). Propagating this key to the child relation, COLLECTION_BIN, would require a three-attribute foreign key. The use of collrun_id simplifies foreign key references, while the original business key is preserved through a candidate key (unique constraint) to ensure each collection run remains uniquely identified.